8 The pre-release material contains information on Pulse Rate and Body Mass Index (BMI). A student is investigating whether there is a relationship between pulse rate and BMI. A **section** of the available data is shown in the table.

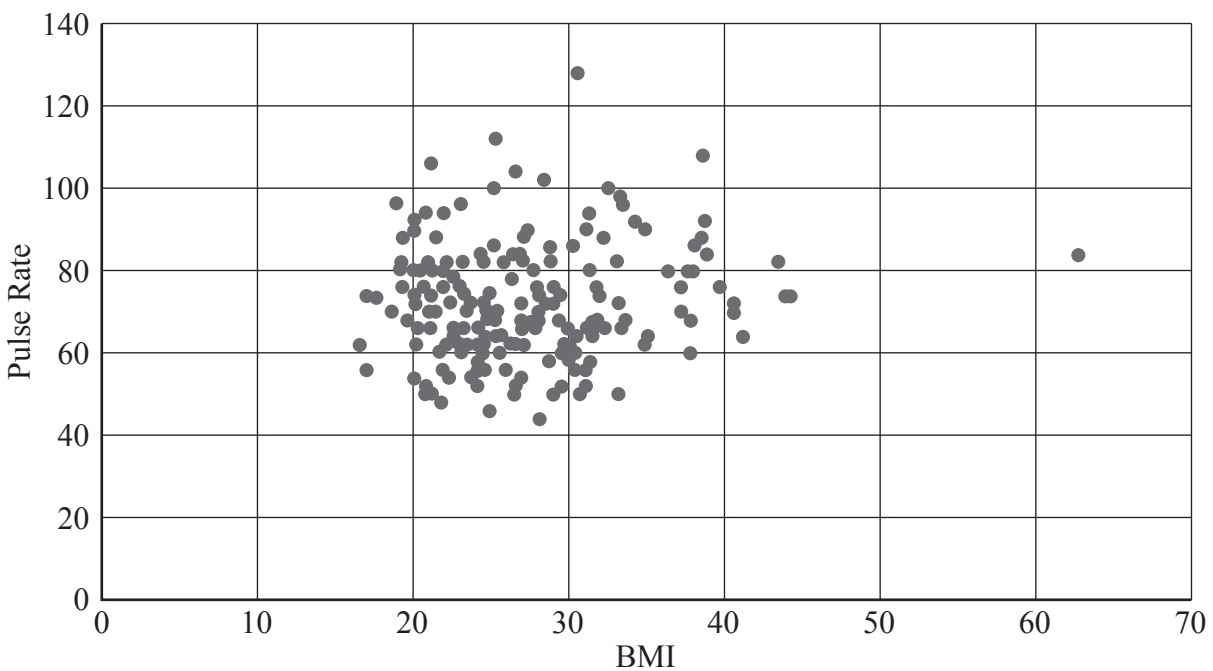
Sex	Age	BMI	Pulse
Male	62	29.54	60
Female	20	23.68	#N/A
Male	17	26.97	72
Male	35	24.7	64
Male	17	20.09	54
Male	85	23.86	54
Female	81	24.04	#N/A

The student decides to draw a scatter diagram.

(a) With reference to the table, explain which data should be cleaned before any analysis takes place. **[1]**

The student cleans the data for BMI and Pulse Rate in the pre-release material and draws a scatter diagram.

Scatter diagram of Pulse Rate against BMI



The student identifies **one** outlier.

(b) On the copy of the scatter diagram in the Printed Answer Booklet, circle this outlier. **[1]**

The student decides to remove this outlier from the data. They then use the LINEST function in the spreadsheet to obtain the following formula for the line of best fit.

$P = 0.29Q + 64.2$,

where P = Pulse Rate and Q = BMI.

They use this to estimate the Pulse Rate of a person with BMI 23.68.
They obtain a value of 71 correct to the nearest whole number.

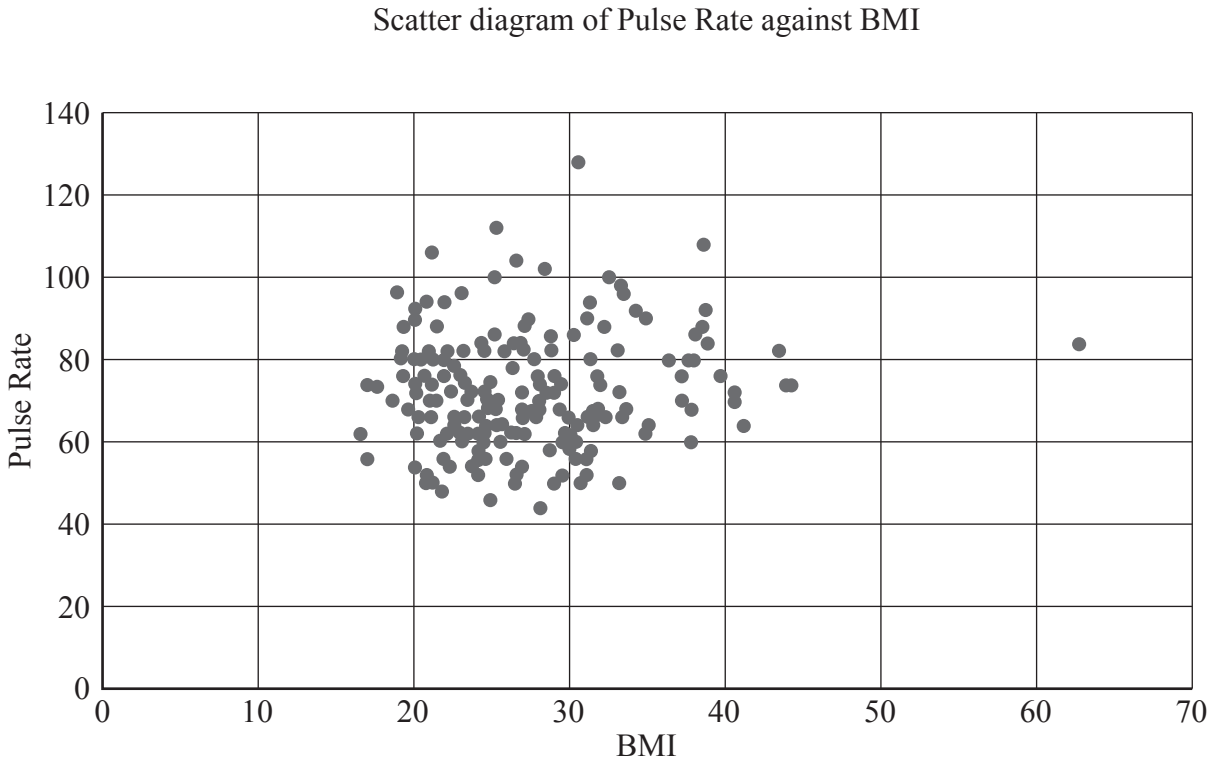
(c) With reference to the scatter diagram, explain whether it is appropriate to use the formula for the line of best fit. **[1]**

It is suggested that all pairs of values where the pulse rate is above 100 should also be cleaned from the data, as they must be incorrect.

(d) Use your knowledge of the pre-release material to explain whether or not all pairs of values with a pulse rate of more than 100 should be cleaned from the data. **[1]**

8(a)	

8(b)


Scatter diagram of Pulse Rate against BMI
Pulse Rate
BMI
0
20
40
60
80
100
120
140
0
10
20
30
40
50
60
70

8(c)

8(d)